Discrete Extramental Time versus Classical Thermodynamic Time

A Formal Confrontation

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Abstract

The Relational Extramental Discrete Clock (RECD) accumulates a signed, depth-rescaled tick driven by Systemic Tau and an operational regime gate. Classical nonequilibrium thermodynamics, by contrast, takes absolute continuous time as an extrinsic, strictly oriented parameter of the evolution semigroup. This note records the mathematical confrontation between these two generators as formalized in Lean 4 (Module CT of `systemic-tau-formal` v0.1.11; software DOI 10.5281/zenodo.21581189). The principal results are: (i) global monotonicity of the RECD accumulant is equivalent to nonnegativity of ticks and fails under positive density of anti-synchronization; (ii) no orientation-respecting bridge maps a path with infinitely often negative increments into a classical discrete clock; (iii) in the stable regime with frozen depth, the accumulant is strictly monotone and bi-Lipschitz with respect to the index and to a uniform immersion mesh; (iv) ordinal production Σ^{RECD} is nonnegative by construction even when the clock itself decreases. Controlled continuum-limit hypotheses, immersion, projection, and estimator-noise envelopes are included at the discharged pathwise level. Empirical series are excluded from all statements. The conclusion is structural: RECD time and classical thermodynamic time are not the same mathematical object.

1. Introduction

1.1 Motivation

Classical continuum thermodynamics writes balance laws on a continuous parameter $t \in \mathbb{R}_{\geq 0}$ that is extrinsic to the state, additive under the evolution semigroup, and oriented so that $dt \geq 0$ along admissible processes. Entropy production

$$\sigma_{\mathrm{cl}} = \frac{d_i S}{dt} \geq 0$$

is a constraint *on processes relative to that parameter*; it does not define t. The same absolute orientation underwrites continuum Markov semigroups, local entropy balance, and the Schnakenberg representation of production on jump networks: the independent variable of the theory is never allowed to run backward.

The Relational Extramental Discrete Clock (RECD) is a different construction. Its tick is generated by an ordinal concordance measure (Systemic Tau τ_s), a piecewise gate g, and a Feigenbaum depth scale δ^{-k}. The resulting accumulant T_n need not be monotone: anti-synchronization forces negative ticks. Null ticks appear on the intermediate nonnegative band. Depth rescaling densifies or contracts the intrinsic step without reference to laboratory seconds. The confrontation is therefore not a matter of notation or units. It concerns whether two generators of "time" share the structural properties that make classical thermodynamic time what it is—monotonicity of the parameter, extrinsic genesis, semigroup additivity, and a production inequality written with respect to that parameter.

This note answers that question at the level of machine-checked mathematics. It does not argue that RECD "replaces" thermodynamic time, nor that laboratory series validate theorems. It shows what is already proved: where the generators agree locally, where they diverge globally, and which bridges (immersion, projection, noise envelopes) are controlled. The methodological commitment is that a structural non-identity, once formalized, does not require empirical illustration to be a mathematical fact.

1.2 Scope

Only results already formalized in Lean under `SystemicTau/` for Module CT are treated as theorems. Conjectures and open interfaces are listed separately and never used as premises. Empirical material (epidemiological series, Holter, finance fixtures, and related operational notebooks) is outside the mathematical claims of this note.

The operative software pin is **v0.1.11** (DOI 10.5281/zenodo.21581189), including the ContLim+noise residual pack. The full design document of the module is `docs/RECD_vs_Thermodynamic_Time.md`; the present note is a self-contained scientific condensation of the discharged core.

1.3 Epistemic discipline

Claims are stratified. Operational definitions (gate law, tick formula, mesh conventions) are protocol choices, not theorems about nature. Theorems are arithmetic and order-theoretic consequences of those definitions, machine-checked with 0 `sorry`. Physical interpretations and ontological readings are not premises of proofs. Classical continuum thermodynamics is treated as an *external comparison theory*: no thermodynamic axiom is added to the Lean library, and σ_{cl} is not identified with ordinal production Σ^{RECD}.

This discipline is enforced in the source (`docs/EPISTEMIC_LABELS.md`) and in the Lean modules by construction: research residual statements are registered as `Props`, not as `axioms` (see `RECD_ResearchOpen.lean`).

2. Two generators of time

2.1 The RECD generator

Fix a sequence of Systemic Tau samples $(\tau_s^{(n)})_{n\in\mathbb{N}}$ with values in $\mathbb{Q}$ (the Lean skeleton) and a depth schedule $(k_n)_{n\in\mathbb{N}}$. The operational Feigenbaum constant $\delta > 1$ appears through the positive rational scale δ^{-k} (`deltaInvPow`). Thresholds are fixed as in Lean: $\tau_{\text{st}} = 1/2$, $\tau_{\text{ch}} = 41/100$.

Gate. The reference gate $g\colon \mathbb{Q} \to \mathbb{Q}$ (Lean `gate`, `RECD.lean`) is piecewise:

$$g(\tau) = \begin{cases} +1 & \text{if } \tau \geq \tau_{\text{st}}, \\ \dfrac{\delta-1}{\delta} \cdot \dfrac{\tau_{\text{ch}} - |\tau|}{\tau_{\text{ch}}} & \text{if } |\tau| < \tau_{\text{ch}}, \\ -1 & \text{if } \tau \leq -\tau_{\text{ch}}, \\ 0 & \text{otherwise.} \end{cases}$$

The law is **not** globally odd or even. On the open chaotic band, g depends only on $|\tau|$; on the nonnegative chaotic half-band it is antitone; stable forces $g = +1$; anti-sync forces $g = -1$; the nonnegative intermediate band $\tau_{\text{ch}} \leq \tau < \tau_{\text{st}}$ forces $g = 0$. These facts are discharged (`gate_of_stable`, `gate_of_antiSync`, `gate_neg_iff_antiSync`, etc.).

Tick and accumulation. The unit tick and accumulant are

$$\Delta T_n = g(\tau_s^{(n)}) \cdot \delta^{-k_n} \cdot \left|\tau_s^{(n)}\right|,$$
$$T_0 = 0, \qquad T_{n+1} = T_n + \Delta T_n.$$

In Lean: `recdTick_unit`, `recdAccum` (`RECD.lean`, `RECD_Monotonicity.lean`). The map $n \mapsto T_n$ is a real (here rational) sequence; it need not be monotone.

Regime partition. Classification into stable / chaotic / anti-sync / intermediate is operational (`Thresholds.lean`). Sign and magnitude of ΔT_n are regime-dependent through g and $|\tau_s|$; scale is depth-dependent through δ^{-k_n}.

2.2 The classical thermodynamic generator

As comparison object we fix the standard continuum package:

1. An absolute continuous parameter $t \in \mathbb{R}_{\geq 0}$, extrinsic to the state.
2. An evolution semigroup $(P_t)_{t\geq 0}$ with $P_{t+s} = P_t \circ P_s$, $P_0 = \mathrm{Id}$.
3. Strict orientation of the parameter: physical clocks advance only in the $+t$ direction; reverse trajectories, when considered, are different processes, not negative increments of the same absolute t.
4. Internal entropy production $\sigma_{\text{cl}}(t) = \frac{d_i S}{dt} \geq 0$ along admissible evolutions (Clausius–Duhem /

local balance; Schnakenberg form on jump networks).

Under these axioms, $\sigma_{\mathrm{cl}} \geq 0$ constrains processes *relative to* t; it does not generate t. Classical t is independent of Kendall-type ordinal concordance among measured channels.

In the discrete Lean confrontation, classical orientation is represented by the path property `IsClassicalOriented`: every successive increment is nonnegative (`RECD_Oriented.lean`). This is a disciplined stand-in for $dt \geq 0$, not a full continuum semigroup.

2.3 Structural incompatibilities

The following axes are structural, not empirical.

Axis	RECD generator T_n	Classical thermodynamic t
Type	Countable ticks; step path in the accumulant	Continuous (or dense) parameter of a flow/semigroup
Monotonicity	ΔT_n may be > 0, $= 0$, or < 0	t nondecreasing; negative dt not an admissible clock increment
Genesis	Generated by τ_s and gate g	Extrinsic absolute parameter of the balance laws
Scale	Depth multiplies the tick by δ^{-k}	No Feigenbaum depth factor in dt
Null ticks	Possible when $g = 0$ or $\tau_s = 0$	dt does not freeze by ordinal band membership
Entropy carrier	Ordinal production Σ^{RECD} from ticks	$\sigma_{\mathrm{cl}} \geq 0$ is primitive

Immediate corollary. The identity of parameters "$T_n \equiv t$" is not an isomorphism of generators: one admits negative increments, null ticks, and δ-renormalization; the other does not. This is already elementary from the gate law: constant anti-sync $\tau_s^{(n)} \leq -\tau_{\mathrm{ch}}$ yields $\Delta T_n < 0$ for all n with $|\tau_s| > 0$.

3. Main formal results

Throughout, statements marked as theorems are discharged in Lean (0 `sorry`). We omit long proofs and record the content with Lean anchors.

3.1 CT-3 — Monotonicity criteria

Theorem (CT-3; `RECD_Monotonicity.lean`).
For a fixed finite horizon N, the following are equivalent:

1. $T_{n+1} \geq T_n$ for all $n < N$;
2. $\Delta T_n \geq 0$ for all $n < N$;
3. $g(\tau_s^{(n)}) \geq 0$ for all $n < N$ with $\tau_s^{(n)} \neq 0$.

Anchors: `recdAccum_monotone_iff_ticks`, `recdAccum_monotone_iff_gate`, `recdTick_unit_nonneg_iff_gat`

Theorem (gate support). Under the reference law, $g(\tau) < 0$ if and only if $\tau \leq -\tau_{\mathrm{ch}}$.
Anchors: `gate_neg_iff_antiSync`, `gate_nonneg_iff_gt_neg_chaos`.

Corollary. If $\tau_s^{(n)} > -\tau_{\mathrm{ch}}$ for all $n < N$, then $(T_n)_{n \leq N}$ is nondecreasing (`recdAccum_monotone_of_gt_neg_chaos`).

Quantitative recovery (`RECD_Variation.lean`).
If $|\tau_s^{(n)}| \le B$ and the number of anti-sync visits up to N is at most M, then reverse mass satisfies $\sum_{n<N} \Delta T_n^- \le B \cdot M$, hence $T_N \ge -B \cdot M$. Depth need not be capped separately: $\delta^{-k} \le 1$ for every k (`deltaInvPow_le_one`).
Anchors: `negVarAccum_le_antiSync_mass`, `recdAccum_ge_neg_of_antiSync_le`, `recdAccum_ge_neg_M_unit`.

3.2 CT-2 — Failure under anti-sync density

Theorem (negative ticks). If $\tau \le -\tau_{\mathrm{ch}}$ and $|\tau| \ge c > 0$, then $\Delta T \le -c \cdot \delta^{-k} < 0$ (`recdTick_unit_antiSync_le_neg`).

Definition. Infinitely often anti-sync (`InfOftenAntiSync`): for every N there exists $n \ge N$ with anti-sync and amplitude floor. Positive lower density (`HasAntiSyncLowerDensity`) forces InfOften.

Theorem (CT-2 elementary).
If anti-sync occurs infinitely often with amplitude floor $c > 0$, then $T = \mathrm{recdAccum}$ is **not eventually nondecreasing**. The same conclusion follows from positive lower density of anti-sync.
Anchors: `not_eventually_nondecreasing_of_InfOftenAntiSync`, `not_eventually_nondecreasing_of_lowe` (`RECD_CT2.lean`).

Quantitative reverse mass. Under frozen depth, density, and amplitude floor, reverse variation grows at least linearly
(`negVarAccum_ge_linear_of_density`).

Witnesses. Constant anti-sync (density 1) and period-2 anti-sync/stable (density 1/2): the latter satisfies $T_{2m} = 0$, $T_{2m+1} = -3/4$ under the standard sample and is neither eventually nondecreasing nor eventually nonincreasing
(`period2AntiSync_not_eventually_monotone`, `period2AntiSync_not_eventually_nondecreasing`, `period2AntiSync_not_eventually_nonincreasing`, `recdAccum_period2_even`, `recdAccum_period2_odd`).

Immersion form. On a cell with $\Delta T_n < 0$, the affine immersion is strictly antitone in the mesh parameter
(`immerseAffine_strict_anti_of_neg_tick`).

This is the elementary half of global non-equivalence: classical clock orientation fails as soon as anti-sync recurs with positive mass. The orientation-bridge obstruction is stated next.

3.3 Orientation — no classical bridge for reverse paths

Definitions (`RECD_Oriented.lean`).

- `IsClassicalOriented`: $\forall n,\ T_n \le T_{n+1}$.
- `OrientationRespecting`: for all $n \ge N_0$, a negative source increment of T_{src} forces a negative target increment of T_{tgt}.
- `InfOftenNegInc`: infinitely often strictly negative path increments.

Theorem (local obstruction). A classical oriented target cannot respect a negative source increment at the same index
(`classical_not_respect_neg_inc`, `not_orientationRespecting_of_classical_at`).

Theorem (OP-CT-8 elementary). If the source has infinitely often negative increments and the target is classical (or only eventually classical), then there is no N_0 such that the correspondence is orientation-respecting after N_0

(`no_eventual_orientationRespecting_classical`, `no_orientationRespecting_eventually_classical`).

Corollary for RECD. Constant anti-sync and period-2 RECD walks are `InfOftenNegInc`; no classical clock respects their orientation under identity indexing (`constAntiSync_no_classical_orientation`, `period2_no_classical_orientation`, `constAntiSync_InfOften` `period2_InfOftenNegInc`).

Monoidal residual (`RECD_Monoidal.lean`). Discrete monoid morphisms $\mathbb{N} \to \mathbb{N}$ are multiplication by a constant; there is no oriented monoid bridge from a reverse-oriented RECD walk into a classical clock (`no_OrientedMonoidBridge_classical`). Tensor of classical clocks remains classical (`pathTensor_classical`). Full Mathlib CategoryTheory monoidal functors on continuum path spaces remain open; the discrete obstruction is discharged.

Reading. These theorems formalize the claim that one cannot reindex away sustained reverse orientation while keeping a classical clock. They do **not** assert a full categorical non-existence theorem for continuum semigroup intertwining functors; that residual is open (OP-CT-8 continuum half).

3.4 CT-1 — Local bi-Lipschitz equivalence (stable regime)

Hypotheses (stable skeleton). On a discrete interval of length N: $\tau_s^{(n)} \geq \tau_{\text{st}}$ (hence $g \equiv +1$); depth frozen at $k_\star$; immersion mesh uniform with step $h > 0$.

Theorem (tick law). $\Delta T_n = \delta^{-k_\star}\tau_s^{(n)} \geq \delta^{-k_\star}\tau_{\text{st}} > 0$ (`recdTick_unit_of_stable`, `recdTick_unit_pos_of_stable`).

Theorem (strict mono and order embedding). $i \leq j \Rightarrow T_i \leq T_j$; $i < j \Rightarrow T_i < T_j$ (`recdAccum_strictMono_of_stable`, `recdAccum_le_of_le_stable`, `recdAccum_lt_of_lt_stable`).

Theorem (sandwich and bi-Lipschitz). If $m \leq \tau_s^{(n)} \leq M$ with $m > 0$, then

$$n \cdot \delta^{-k_\star} m \;\leq\; T_n \;\leq\; n \cdot \delta^{-k_\star} M,$$

and $n \mapsto T_n$ is bi-Lipschitz with constants $\ell = \delta^{-k_\star} m$, $L = \delta^{-k_\star} M$. Mesh form, with $\ell = \delta^{-k_\star} m/h$ and $L = \delta^{-k_\star} M/h$:

$$\ell \cdot (s_b - s_a) \;\leq\; T_b - T_a \;\leq\; L \cdot (s_b - s_a).$$

Anchors: `recdAccum_stable_sandwich`, `recdAccum_stable_biLipschitz`, `recdAccum_stable_biLipschitz_m` (`RECD_CT1.lean`).

Sample. Constant stable $\tau \equiv 3/4$, depth 0: $T_n = n \cdot (3/4)$ exactly (`recdAccum_constStable_exact`).

Interpretation. Under stable domination and frozen depth, the RECD accumulant is locally equivalent—order-isomorphic and bi-Lipschitz on compact horizons—to an oriented classical discrete clock (index time or uniform mesh time). This is *conditional* local equivalence, not global identity of generators.

3.5 Ordinal production $\Sigma^{\text{RECD}} \geq 0$ by construction

Definition. Forward and backward ordinal masses:

$$\Delta T_n^+ = \max(\Delta T_n, 0), \qquad \Delta T_n^- = \max(-\Delta T_n, 0).$$

For asymmetry parameter $\alpha \in [0,1]$,

$$\Sigma_n^{\mathrm{RECD}} = \Delta T_n^+ + \alpha\,\Delta T_n^-, \qquad\qquad \Sigma_N^{\mathrm{RECD}} = \sum_{n<N} \Sigma_n^{\mathrm{RECD}}.$$

(The plus combination is essential: when $\Delta T_n < 0$, $\Sigma_n = \alpha \Delta T_n^- \geq 0$.)

Theorem (CT-4A). For every RECD path and every $\alpha \in [0,1]$, $\Sigma_N^{\mathrm{RECD}} \geq 0$ (`ordinalSigmaAccum_nonneg`, `RECD_OrdinalEntropy.lean`).

Theorem (CT-4B). If $g \geq 0$ on the whole horizon, then $\Delta T_n^- = 0$, hence $\Sigma_N^{\mathrm{RECD}} = T_N - T_0$ (`ordinalSigmaAccum_eq_recdAccum_of_gate_nonneg`).

Mesh density (`RECD_Immersion.lean`). On the uniform mesh $s_n = n \cdot h$, cell density $\sigma_{\mathrm{RECD}} = \Sigma(\Delta T_n; \alpha)/h$ is nonnegative on every open cell (`sigmaRECD_nonneg`). Under $g \geq 0$, σ_{RECD} equals the immersion slope (`sigmaRECD_eq_immerseSlope_of_gate_nonneg`).

Structural split. Production nonnegativity does **not** force clock monotonicity. The constant anti-sync sample has T_n strictly decreasing while $\Sigma_N^{\mathrm{RECD}} > 0$ for $\alpha > 0$ (`ordinal_vs_clock_split_sample`). Classical absolute time forbids this split: the parameter t itself cannot reverse while production remains well-defined along t.

Epistemic note. $\Sigma^{\mathrm{RECD}} \geq 0$ is pure arithmetic on ticks. Identifying it with Clausius entropy production is a modelling hypothesis, not a theorem. Proposition CT-4C (two-sided comparison with Schnakenberg σ_{cl}) remains a conjecture (`CT4C_SchnakenbergConjecture` in `RECD_ResearchOpen.lean`).

3.6 Immersion, variation, projection, and estimator noise

Immersion ι. Uniform mesh $s_n = n \cdot h$; node values `immerseNode`; affine interpolation `immerseAffine` with slope $\Delta T_n/h$ (`RECD_Immersion.lean`). Continuity of the piecewise-affine path on each cell is elementary. Full $C([0,\infty), \mathbb{R})$ as a Mathlib path type is optional packaging, not required for the discrete theorems.

Bounded variation (`RECD_BV.lean`). Total variation

$$\mathrm{TV}_N = \sum_{n<N} |\Delta T_n| = \sum_{n<N} \Delta T_n^+ + \sum_{n<N} \Delta T_n^-, \qquad |T_N| \leq \mathrm{TV}_N.$$

Cell Lipschitz identity: $|\iota(\theta_2) - \iota(\theta_1)| = |\theta_2 - \theta_1| \cdot |\Delta T|$. Period-2 sample: $\mathrm{TV}_N = N \cdot \frac{3}{4}$ while T oscillates boundedly—variation grows, clock does not.

Projection π. Combinatorial Kendall and mean pairwise Systemic Tau on windows are discharged (`kendallTau`, `systemicTauWindow`, `projectPath` in `Basic.lean` / `RECD_Projection.lean`). Projection is operational protocol (window W, sampling, depth rule); it is not injective in general. Statistical consistency in probability of the estimator is **not** claimed.

Estimator noise. Structure `EstimatorNoise` with $\hat{\tau} = \tau_{\mathrm{true}} + \varepsilon$ and envelope `NoiseBounded` ($|\varepsilon_n| \leq \eta$ on a horizon). Stable band is preserved under margin $\tau_{\mathrm{true}} \geq \tau_{\mathrm{st}} + \eta$ (`estimator_preserves_stable`). CT-1 residual (`RECD_CT1_Noise.lean`): noise-robust mono/strict mono under hat-stable; single-tick Lip $|\Delta\widehat{T} - \Delta T| = \delta^{-k}|\varepsilon|$; accum gap

$$\left|\widehat{T}_N - T_N\right| \;\leq\; N \cdot \delta^{-k} \cdot \eta$$

(`recdAccum_noise_bound`). Discrete shift/inc cocycle and intertwining obstruction under InfOften negative increments are discharged; continuum semigroup P_t intertwining is not.

ContLim + noise (`RECD_ContLim.lean`, `RECD_ContLimFull.lean`, `RECD_ContLim_Noise.lean`). A controlled continuum limit is a regime in which the immersed RECD accumulant converges to a nondecreasing absolutely continuous clock with nonnegative density. The Lean development packages sufficient hypotheses H1–H4 at the discrete pathwise level:

- **H1** stable domination ($\hat{\tau} \geq \tau_{\mathrm{st}}$ on the horizon);
- **H2** frozen (bounded) depth;
- **H3** regular sampling flag (operational protocol placeholder);
- **H4** positive mass lower bound on $|\tau|$.

Discharged consequences include: equi-Lipschitz immersion under stable frozen depth and amplitude upper bound; const-stable exact linear continuum shape with Real `Tendsto atTop`; uniform envelope and equicontinuous modulus on compact horizons; projected ContLim under H1–H4 on π. Under `NoiseBounded`: degraded/inflated band $\tau_{\mathrm{st}} - \eta \leq \hat{\tau} \leq M + \eta$; the same accum gap bound; equi-Lip immersion of the hat path with modulus $\delta^{-k}(M+\eta)$; small-η mono of estimated immersion on stable cells; packaged as `ContLimNoisePackage`.

Obstruction. InfOften anti-sync with amplitude floor forbids any nondecreasing continuum limit of the immersed accumulant (`contLim_obstruction_of_InfOftenAntiSync`). H1-type control is essentially necessary for classical orientation recovery. The continuum limit is therefore conditional, not universal—consistent with CT-2 and OP-CT-8.

4. Structural consequences

4.1 The RECD clock can recede; classical thermodynamic time cannot

From CT-3 and CT-2: negative ticks exist precisely on the anti-sync half-line $\tau \leq -\tau_{\mathrm{ch}}$ (with $\tau \neq 0$). When such visits recur infinitely often, T_n is not eventually nondecreasing. Classical t, by the comparison package and by `IsClassicalOriented`, admits no negative increments. Therefore global identification of T_n with t fails on any trajectory class that sustains anti-synchronization.

4.2 Ordinal production remains nonnegative when the clock is not monotone

CT-4A separates two objects that classical thermodynamics fuses through a single parameter:

Object	Behaviour under anti-sync
Clock T_n	May decrease
Production Σ_N^{RECD}	Stays ≥ 0 for every $\alpha \in [0,1]$

Under $g \geq 0$, the two coincide (CT-4B). Under mixed signs, production measures oriented ordinal reorganization mass (forward plus weighted reverse), not the net displacement of the clock. This is the structural content of the “ordinal second law” as a definitional inequality, not as a physical law of nature.

4.3 Precise conditions for local classical behaviour

The RECD generator behaves *locally* as an oriented classical discrete clock when **all** of the following hold on a horizon:

1. **Stable domination:** $\tau_s^{(n)} \geq \tau_{\text{st}}$ (hence $g \equiv +1$);
2. **Frozen depth:** $k_n = k_\star$;
3. **Amplitude control:** $0 < m \leq \tau_s^{(n)} \leq M$ (for bi-Lipschitz constants);
4. **Declared mesh** (if immersion is used): uniform $h > 0$.

Then CT-1 applies: strict mono, sandwich, bi-Lipschitz vs index and mesh. With estimator noise, the same conclusions hold for $\widehat{T}$ under a margin that preserves the stable band, with an additive gap linear in $N \cdot \eta \cdot \delta^{-k}$.

ContLim scaffolding adds equi-Lipschitz and equicontinuous moduli under H1–H4, and noise-stable variants under small η. The obstruction theorems show that dropping (1) in a recurrent way is fatal for classical orientation: InfOften anti-sync alone forbids eventual nondecreasing recovery and forbids orientation-respecting bridges. Dropping (2) without further control leaves depth free to collapse ticks by δ^{-k}; the discrete reverse-mass bound of CT-3 quantitative still applies under amplitude control, but bi-Lipschitz constants against a fixed mesh must be re-derived if depth varies. None of these statements reinstall global equivalence of generators. Local classicality is a *regime*, not an identity of objects.

4.4 Second-law carriers are not interchangeable

Classical thermodynamics places $\sigma_{\text{cl}} \geq 0$ relative to absolute t. Module CT defines Σ^{RECD} relative to the RECD tick algebra. When $g \geq 0$, CT-4B yields $\Sigma_N^{\text{RECD}} = T_N - T_0$, so production and clock displacement coincide. When anti-sync is present, they diverge. Any modelling claim that "ordinal production *is* thermodynamic entropy production" is therefore at least as strong as CT-4C, which remains open. The formal confrontation does not need CT-4C: non-identity of generators follows from orientation and monotonicity alone.

4.5 Bridges are not identifications

Immersion ι embeds the discrete accumulant into a continuous path so that continuum language can be applied *without* claiming the generators are identical. Projection π extracts ordinal structure from a continuous multivariate series under a protocol. Composition $\pi \circ \iota$ is not the identity in general. Changing window, mesh, or depth rule changes the RECD path; that dependence is measurement protocol, not a dynamical theorem. The Lean development makes this explicit by typing immersion on rational paths with a declared mesh and by packaging projection and noise as operational structures rather than as equalities of trajectory spaces.

5. What is left open

The following fronts are research-scale. They do **not** undermine the structural confrontation already formalized; they delimit what remains beyond pathwise discrete mathematics.

1. **Full continuum ContLim for random projected processes.** Equicontinuity and node ContLim on rational paths are discharged; weak convergence / Skorokhod topology for general random projected continuous processes is not (`RECD_ContLimFull.lean` residual).
2. **Probabilistic Kendall consistency.** Combinatorial τ_s on finite windows is discharged; consistency in probability of the estimator under mixing assumptions is not.
3. **Continuum semigroup intertwining.** Discrete shift/inc cocycle and orientation obstruction are discharged; intertwining with a continuum Markov semigroup (P_t) is open.

4. **Schnakenberg bridge (CT-4C).** Two-sided comparison of Σ^{RECD} with classical σ_{cl} on a jump network remains a conjecture (`CT4C_SchnakenbergConjecture`). Only the RECD half $\Sigma \geq 0$ is a theorem.
5. **Ordinal large-deviation principle.** Fluctuation theory for $\Sigma_N^{\mathrm{RECD}}/N$ under stationary chaotic τ_s with Feigenbaum depth schedule is open (`OrdinalFluctuationConjecture`). InfOften skeleton is available; rate functions are not.
6. **Nested multi-scale RECD ContLim.** Layered production nonnegativity and orientation obstruction reuse are partial; multi-scale continuum limit is open.
7. **Mathlib categorical monoidal functors.** Discrete monoid/tensor stand-ins are discharged; monoidal functors on continuum path categories are not.

These items are registered honestly in `RECD_ResearchOpen.lean` as interfaces, not axioms.

6. Conclusion

The Relational Extramental Discrete Clock and the absolute time of classical continuum thermodynamics are not the same mathematical object. The difference is structural and already machine-checked:

- **Monotonicity** of the RECD accumulant is equivalent to nonnegativity of ticks and fails under sustained anti-synchronization (CT-3, CT-2).
- **Orientation** cannot be preserved by any eventual bridge from reverse-recurrent RECD walks into a classical discrete clock (OP-CT-8 elementary).
- **Local equivalence** holds in the stable frozen-depth regime as bi-Lipschitz order embedding (CT-1), and survives controlled estimator noise with an explicit gap.
- **Ordinal production** Σ^{RECD} is nonnegative by construction even when the clock recedes (CT-4A/B)—a split classical absolute time does not admit for its own parameter.

Bridges—immersion, projection, ContLim hypotheses, noise envelopes—make the comparison precise without collapsing the generators. What remains open (probabilistic ContLim, Schnakenberg comparison, LDP, continuum monoidal functors) is research, not a gap in the structural non-identity already proved. In particular, strengthening ContLim to random continuous processes or linking Σ^{RECD} to Schnakenberg currents would refine *how* the two theories can be coupled in applications; it would not restore global identity of the time generators themselves.

Module CT therefore supplies a formal, epistemically stratified confrontation: RECD time can reverse; classical thermodynamic time cannot; production can stay nonnegative while the clock does not; and the conditions under which RECD *looks* classical are exact, local, and verified. The confrontation is complete at the structural level demanded by the module's design. Further formal work is optional enrichment of open analytic and probabilistic fronts, not a prerequisite for the non-equivalence theorems recorded here.

Lean formalization status

Formalization under `SystemicTau/` in `systemic-tau-formal` v0.1.11 (Lean 4.14.0, Mathlib v4.14.0): 0 `sorry`, 0 research `axiom`.

Result class	**Lean module(s)**	**Status**
Gate law, thresholds, tick skeleton	`RECD.lean`, `Thresholds.lean`	Machine-checked
CT-3 mono criteria + gate support	`RECD_Monotonicity.lean`	Machine-checked
CT-3 quantitative reverse-mass bound	`RECD_Variation.lean`	Machine-checked
CT-4A/B ordinal production	`RECD_OrdinalEntropy.lean`	Machine-checked
Immersion mesh + σ_{RECD}	`RECD_Immersion.lean`	Machine-checked
CT-1 stable bi-Lipschitz skeleton	`RECD_CT1.lean`	Machine-checked
CT-2 InfOften / density / period-2	`RECD_CT2.lean`	Machine-checked
Immersion BV / Lip / TV	`RECD_BV.lean`	Machine-checked
OP-CT-8 orientation obstruction	`RECD_Oriented.lean`	Machine-checked (discrete)
ContLim H1–H4 scaffold + obstruction	`RECD_ContLim.lean`	Machine-checked (pathwise)
ContLim residual (envelope, equicont)	`RECD_ContLimFull.lean`	Machine-checked (pathwise)
ContLim + noise package	`RECD_ContLim_Noise.lean`	Machine-checked (pathwise)
Projection + estimator noise	`RECD_Projection.lean`	Machine-checked (combinatorial)
CT-1 noise + discrete cocycle	`RECD_CT1_Noise.lean`	Machine-checked
Monoidal residual (discrete)	`RECD_Monoidal.lean`	Machine-checked
Research residual registry	`RECD_ResearchOpen.lean`	Interfaces only (open)

Not machine-checked (and not claimed): continuum Markov P_t intertwining; Schnakenberg CT-4C inequality; ordinal LDP rate functions; probabilistic Kendall consistency; Skorokhod ContLim for general random processes; Mathlib categorical monoidal functors on continuum path spaces.

References

1. Padilla-Villanueva, J. *systemic-tau-formal* v0.1.11. Software. DOI: 10.5281/zenodo.21581189. Concept DOI: 10.5281/zenodo.21516059.

2. Module CT design document: `docs/RECD_vs_Thermodynamic_Time.md` (repository).

3. Epistemic labels: `docs/EPISTEMIC_LABELS.md`.

4. Lean entry points: `lean/SystemicTau/RECD*.lean` as tabulated above.

5. de Groot, S. R. & Mazur, P. *Non-Equilibrium Thermodynamics.* North-Holland / Dover (1962/1984). Continuum entropy balance (external comparison object).

6. Schnakenberg, J. Network theory of microscopic and macroscopic behavior of master equation systems. *Rev. Mod. Phys.* **48**, 571–585 (1976).

7. Classical comparison package used in the note (not formalized in Lean): absolute continuous time as semigroup parameter; Clausius–Duhem / local production $\sigma_{\mathrm{cl}} \geq 0$.